

Md. Danial Dirar

South Badda, Dhaka | (+880) 1818843999 | danielidirar@protonmail.com
linkedin.com/in/md-danial-dirar/ | github.com/Danial-Dirar | ORCID: 0009-0002-4081-4175

EDUCATION

BRAC University

B.Sc. (Hons.) in Computer Science

Dhaka, Bangladesh

June 2022 – Present

- I am currently completing my final semester in the dept. of Computer Science & Engineering.

EXPERIENCE

Arabian Gulf Associates Co. Ltd

Entry Level Software Engineer

Dhaka, Bangladesh

July 2024 – October 2024

- Worked as a part-time junior software engineer.

AFK Productions

Chief Communication Officer

Dhaka, Bangladesh

February 2025 – June 2025

- Led the Communications Department as Head of Communications.

Associates

Steam Store Bangladesh - SSB

Owner

Dhaka, Bangladesh

July 2016 – 2026

- Founded and managed a startup digital company that sells digital products online.

PROJECTS

Machine Learning

- Metagenomic Profiling of Plastic Degrading Microbial Communities – Profiling plastic-degrading microbial communities in urban water bodies using diamond-search and blast algorithms on gene-expression metadata from NHL.
- Ground Water Level Prediction – Building machine learning models to forecast groundwater levels from historical hydrological and environmental data (thesis ongoing)

Artificial Intelligence

- Local Frame Generation with AI – Developed pipelines with local AI models and custom LORA for image generation and further optimizing for custom advert marketing.
- Loan Approval & E-commerce Shipping – Developed AI models for loan approval prediction and e-commerce shipping optimization.

Mobile Application

- DishDash — Android/iOS – Developed a cross-platform mobile app for discovering nearby restaurant offers using realtime location.

Kernel Development

- Haifa OS – Made a modified bootloader and a linux distro that provides a seamless experience for new users of all sorts.

Computer Graphics

- Signal Sweep – A dynamic radar interface scans for nearby Bluetooth devices, visualizes them on a 3D radar display, and provides interactive controls.

Distributed Systems

- Signal Sweep — Implemented a resource-aware distributed machine learning system for which uses a master-worker architecture to train regression models collaboratively across multiple nodes.

Database Project

- Global River Database — Designed and implemented a comprehensive global river database with verification method through hierarchy.

Web Development

- Haifa Intelligence — A portfolio website for our projects at www.haifaintelligence.com

PUBLICATIONS

Engineering hydrological resilience: Quantitative assessment of managed aquifer recharge (MAR) potential utilizing advanced horizontal flow treatment in water stressed regions. *Springer Nature.*

Design and field performance of an integrated horizontal flow treatment managed aquifer recharge system for groundwater rehabilitation in Bangladesh. *Springer Nature.*

SKILLS

Programming Languages: Python, C, C++, C#, Dart, Lua

Tools & Technologies: Git, OpenGL, ArcGIS, Flutter, PyTorch, NestJS, NextJS, ShadCN

LANGUAGE

- Bengali (Native)
- English (Bi-lingual proficiency)
- Spanish (Basic Communication)
- Japanese (Basic Communication)

ACHIEVEMENTS

6th Place, National Programming Contest 2017, Bangladesh — organized by a2i.